IET C-DAC ACTS ATC, Pune

DAY_01

DATE : 21-09-2026

TIME : 6:30 P.M. TO 8 : 30 P.M.

TOPIC_01: LCM AND HCF

1. Find the greatest number that will divide 43, 91 and 183 so as to leave the same remainder in each case.
   A. 4
   B. 7
   C. 9
   D. 13

2. The H.C.F. of two numbers is 23 and the other two factors of their L.C.M. are 13 and 14. The larger of the two numbers is:
   A. 276
   B. 299
   C. 322
   D. 345

3. Six bells commence tolling together and toll at intervals of 2, 4, 6, 8 10 and 12 seconds respectively. In 30 minutes, how many times do they toll together ?
   A. 4
   B. 10
   C. 15
   D. 16

4. Let N be the greatest number that will divide 1305, 4665 and 6905, leaving the same remainder in each case. Then sum of the digits in N is:
   A. 4
   B. 5
   C. 6
   D. 8

5. The greatest number of four digits which is divisible by 15, 25, 40 and 75 is:
   A. 9000
   B. 9400
   C. 9600

D. 9800

6. The product of two numbers is 4107. If the H.C.F. of these numbers is 37, then the greater number is:
   A. 101
   B. 107
   C. 111
   D. 185

7. Three number are in the ratio of 3 : 4 : 5 and their L.C.M. is 2400. Their H.C.F. is:
   A. 40
   B. 80
   C. 120
   D. 200

8. Three number are in the ratio of 3 : 4 : 5 and their L.C.M. is 2400. Their H.C.F. is:
   A. 40
   B. 80
   C. 120
   D. 200

9. The product of two numbers is 2028 and their H.C.F. is 13. The number of such pairs is:
   A. 1
   B. 2
   C. 3
   D. 4

10. The least multiple of 7, which leaves a remainder of 4, when divided by 6, 9, 15 and 18 is:
    A. 74
    B. 94
    C. 184
    D. 364

11. Find the lowest common multiple of 24, 36 and 40.
    A. 120
    B. 240

C. 360
D. 480

12. The least number which should be added to 2497 so that the sum is exactly divisible by 5, 6, 4 and 3 is:
A. 3
B. 13
C. 23
D. 33

13. Reduce 128352/238368 to its lowest terms.
A. 3/4
B. 5/13
C. 7/13
D. 9/13

14. The least number which when divided by 5, 6 , 7 and 8 leaves a remainder 3, but when divided by 9 leaves no remainder, is:

A. 1677
B. 1683
C. 2523
D. 3363

15. A, B and C start at the same time in the same direction to run around a circular stadium. A completes a round in 252 seconds, B in 308 A, B and C start at the same time in the same direction to run around a circular stadium. A completes a round in 252 seconds, B in 308 seconds and c in 198 seconds, all starting at the same point. After what time will they again at the starting point ?

A. 26 minutes and 18 seconds
B. 42 minutes and 36 seconds
C. 45 minutes
D. 46 minutes and 12 seconds

TOPIC_02 : RATIO AND PROPORTION

1. A and B together have Rs. 1210. If of A's amount is equal to of B's amount, how much amount does B have?
   A. Rs. 460
   B. Rs. 484
   C. Rs. 550
   D. Rs. 664

2. Two numbers are respectively 20% and 50% more than a third number. The ratio of the two numbers is:
   A. 2 : 5
   B. 3 : 5
   C. 4 : 5
   D. 6 : 7

3. A sum of money is to be distributed among A, B, C, D in the proportion of 5 : 2 : 4 : 3. If C gets Rs. 1000 more than D, what is B's share?
   A. Rs. 500
   B. Rs. 1500
   C. Rs. 2000
   D. None of these

4. Seats for Mathematics, Physics and Biology in a school are in the ratio 5 : 7 : 8. There is a proposal to increase these seats by 40%, 50% and 75% respectively. What will be the ratio of increased seats?
   A. 2 : 3 : 4
   B. 6 : 7 : 8
   C. 6 : 8 : 9
   D. None of these
5. In a mixture 60 litres, the ratio of milk and water 2 : 1. If this ratio is to be 1 : 2, then the quantity of water to be further added is:
   A. 20 litres
   B. 30 litres
   C. 40 litres
   D. 60 litres

6. The ratio of the number of boys and girls in a college is 7 : 8. If the percentage increase in the number of boys and girls be 20% and 10% respectively, what will be the new ratio?
   A. 8 : 9
   B. 17 : 18
   C. 21 : 22
   D. Cannot be determined

7. Salaries of Ravi and Sumit are in the ratio 2 : 3. If the salary of each is increased by Rs. 4000, the new ratio becomes 40 : 57. What is Sumit's salary?
   A. Rs. 17,000
   B. Rs. 20,000
   C. Rs. 25,500
   D. Rs. 38,000

8. If 0.75 : x :: 5 : 8, then x is equal to:
   A. 1.12
   B. 1.2
   C. 1.25
   D. 1.30

9. The sum of three numbers is 98. If the ratio of the first to second is 2 :3 and that of the second to the third is 5 : 8, then the second number is:
   A. 20
   B. 30
   C. 48
   D. 58

10. If Rs. 782 be divided into three parts, proportional to : : , then the first par
   A. Rs. 182
   B. Rs. 190
   C. Rs. 196
   D. Rs. 204

11. The salaries A, B, C are in the ratio 2 : 3 : 5. If the increments of 15%, 10% and 20% are allowed respectively in their salaries, then what will be new ratio of their salaries?
    A. 3 : 3 : 10
    B. 10 : 11 : 20
    C. 23 : 33 : 60
    D. Cannot be determined

12. If 40% of a number is equal to two-third of another number, what is the ratio of first number to the second number?
    A. 2 : 5
    B. 3 : 7
    C. 5 : 3
    D. 7 : 3

13. fourth proportional to 5, 8, 15 is:
    A. 18
    B. 24
    C. 19
    D. 20

14. Two number are in the ratio 3 : 5. If 9 is subtracted from each, the new numbers are in the ratio 12 : 23. The smaller number is:
    A. 27
    B. 33
    C. 49
    D. 55

15. In a bag, there are coins of 25 p, 10 p and 5 p in the ratio of 1 : 2 : 3. If there is Rs. 30 in all, how many 5 p coins are there?
    A. 50
    B. 100
    C. 150
    D. 200

DAY_02

DATE : 22-09-2026

TIME : 6:30 P.M. TO 8 : 30 P.M.

TOPIC : PARTNERSHIP

1. A and B invest in a business in the ratio 3 : 2. If 5% of the total profit goes to charity and A's share is Rs. 855, the total profit is:
   a. Rs. 1425
   b. Rs. 1500
   c. Rs. 1537.50
   d. Rs. 1576

2. A, B and C jointly thought of engaging themselves in a business venture. It was agreed that A would invest Rs. 6500 for 6 months, B, Rs. 8400 for 5 months and C, Rs. 10,000 for 3 months. A wants to be the working member for which, he was to receive 5% of the profits. The profit earned was Rs. 7400. Calculate the share of B in the profit.
   a. Rs. 1900
   b. Rs. 2660
   c. Rs. 2800
   d. Rs. 2840

3. A, B and C enter into a partnership in the ratio : : . After 4 months, A increases his share 50%. If the total profit at the end of one year be Rs. 21,600, then B's share in the profit is:
   a. Rs. 2100
   b. Rs. 2400
   c. Rs. 3600
   d. Rs. 4000

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4. A, B, C subscribe Rs. 50,000 for a business. A subscribes Rs. 4000 more than B and B Rs. 5000 more than C. Out of a total profit of Rs. 35,000, A receives:
   a. Rs. 8400
   b. Rs. 11,900
   c. Rs. 13,600
   d. Rs. 14,700
5. Three partners shared the profit in a business in the ratio 5 : 7 : 8. They had partnered for 14 months, 8 months and 7 months respectively. What was the ratio of their investments?
   a. 5 : 7 : 8
   b. 20 : 49 : 64
   c. 38 : 28 : 21
   d. None of these

6. A starts business with Rs. 3500 and after 5 months, B joins with A as his partner. After a year, the profit is divided in the ratio 2 : 3. What is B's contribution in the capital?
   a. Rs. 7500
   b. Rs. 8000
   c. Rs. 8500
   d. Rs. 9000

7. A and B entered into partnership with capitals in the ratio 4 : 5. After 3 months, A withdrew of his capital and B withdrew of his capital. The gain at the end of 10 months was Rs. 760. A's share in this profit is:
   a. Rs. 330
   b. Rs. 360
   c. Rs. 380
   d. Rs. 430

8. A and B started a partnership business investing some amount in the ratio of 3 : 5. C joined then after six months with an amount equal to that of B. In what proportion should the profit at the end of one year be distributed among A, B and C?
   a. 3 : 5 : 2
   b. 3 : 5 : 5

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c. 6 : 10 : 5
d. Data inadequate

9. A, B, C rent a pasture. A puts 10 oxen for 7 months, B puts 12 oxen for 5 months and C puts 15 oxen for 3 months for grazing. If the rent of the pasture is Rs. 175, how much must C pay as his share of rent?
   a. Rs. 45
   b. Rs. 50
   c. Rs. 55
   d. Rs. 60

10. A and B started a business in partnership investing Rs. 20,000 and Rs. 15,000 respectively. After six months, C joined them with Rs. 20,000. What will be B's share in total profit of Rs. 25,000 earned at the end of 2 years from the starting of the business?
    a. Rs. 7500
    b. Rs. 9000
    c. Rs. 9500
    d. Rs. 10,000

11. Simran started a software business by investing Rs. 50,000. After six months, Nanda joined her with a capital of Rs. 80,000. After 3 years, they earned a profit of Rs. 24,500. What was Simran's share in the profit?
    a. Rs. 9,423
    b. Rs. 10,250
    c. Rs. 12,500
    d. Rs. 10,500

12. A began a business with Rs. 85,000. He was joined afterwards by B with Rs. 42,500. For how much period does B join, if the profits at the end of the year are divided in the ratio of 3 : 1?
    a. 4 months

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b. 5 months
c. 6 months
d. 8 months

13. Aman started a business investing Rs. 70,000. Rakhi joined him after six months with an amount of Rs.. 1,05,000 and Sagar joined them with Rs. 1.4 lakhs after another six months. The amount of profit earned should be distributed in what ratio among Aman, Rakhi and Sagar respectively, 3 years after Aman started the business?
    a. 7 : 6 : 10
    b. 12 : 15 : 16
    c. 42 : 45 : 56
    d. Cannot be determined

14. Arun, Kamal and Vinay invested Rs. 8000, Rs. 4000 and Rs. 8000 respectively in a business. Arun left after six months. If after eight months, there was a gain of Rs. 4005, then what will be the share of Kamal?
    a. Rs. 890
    b. Rs. 1335
    c. Rs. 1602
    d. Rs. 1780

TOPIC : PERCENTAGE

1. A batsman scored 110 runs which included 3 boundaries and 8 sixes. What percent of his total score did he make by running between the wickets?

    E. 45%
    F. 500/11
    G. 330/11
    H. 55%

---

2. Two students appeared at an examination. One of them secured 9 marks more than the other and his marks was 56% of the sum of their marks. The marks obtained by them are:

A. 39, 30

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B. 41, 32
C. 42, 33
D. 43, 34

---

3. A fruit seller had some apples. He sells 40% apples and still has 420 apples. Originally, he had:

A. 588 apples
B. 600 apples
C. 672 apples
D. 700 apples

---

4. What percentage of numbers from 1 to 70 have 1 or 9 in the unit's digit?

A. 1
B. 14
C. 20
D. 21

---

5. If A = *x*% of *y* and B = *y*% of *x*, then which of the following is true?
   A. A is smaller than B.
   B. A is greater than B
   C. Relationship between A and B cannot be determined.
   D. If *x* is smaller than *y*, then A is greater than B.
   E. None of these

---

6. If 20% of *a* = *b*, then *b*% of 20 is the same as:

A. 4% of *a*
B. 5% of *a*
C. 20% of *a*
D. None of these

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7. In a certain school, 20% of students are below 8 years of age. The number of students above 8 years of age is $\frac{2}{3}$ of the number of students of 8 years of age which is 48. What is the total number of students in the school?
   A. 72
   B. 80
   C. 120
   D. 150
   E. 100

---

8. Two numbers A and B are such that the sum of 5% of A and 4% of B is two-third of the sum of 6% of A and 8% of B. Find the ratio of A : B.
   A. 2 : 3
   B. 1 : 1
   C. 3 : 4
   D. 4 : 3

---

9. A student multiplied a number by 3/5 instead of 5/3 What is the percentage error in the calculation?
   A. 34%
   B. 44%
   C. 54%
   D. 64%

---

10. In an election between two candidates, one got 55% of the total valid votes, 20% of the votes were invalid. If the total number of votes was 7500, the number of valid votes that the other candidate got, was:
    A. 2700
    B. 2900
    C. 3000
    D. 3100

.

11. Three candidates contested an election and received 1136, 7636 and 11628 votes respectively. What percentage of the total votes did the winning candidate get?

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A. 57%
B. 60%
C. 65%
D. 90%

---

12. Two tailors X and Y are paid a total of Rs. 550 per week by their employer. If X is paid 120 percent of the sum paid to Y, how much is Y paid per week?
A. Rs. 200
B. Rs. 250
C. Rs. 300
D. None of these

---

13. Gauri went to the stationers and bought things worth Rs. 25, out of which 30 paise went on sales tax on taxable purchases. If the tax rate was 6%, then what was the cost of the tax free items?
A. Rs. 15
B. Rs. 15.70
C. Rs. 19.70
D. Rs. 20

---

14. Rajeev buys good worth Rs. 6650. He gets a rebate of 6% on it. After getting the rebate, he pays sales tax @ 10%. Find the amount he will have to pay for the goods.
A. Rs. 6876.10
B. Rs. 6999.20
C. Rs. 6654
D. Rs. 7000

---

15. The population of a town increased from 1,75,000 to 2,62,500 in a decade. The average percent increase of population per year is:
A. 4.37%
B. 5%
C. 6%
D. 8.75%

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DAY_03

DATE : 23-09-2026

IET C-DAC ACTS ATC, Pune

TIME : 6:30 P.M. TO 8 : 30 P.M.

TOPIC : NUMBER SYSTEM

1. Which one of the following is not a prime number?

   a. 31
   b. 61
   c. 71
   d. 91

---

2. $(112 \times 5^4) = ?$

   a. 67000
   b. 70000
   c. 76500
   d. 77200

---

3. It is being given that $(2^{32} + 1)$ is completely divisible by a whole number. Which of the following numbers is completely divisible by this number?

   a. $(2^{16} + 1)$
   b. $(2^{16} - 1)$
   c. (7×2^{23})
   d. $(2^{96} + 1)$

---

4. What least number must be added to 1056, so that the sum is completely divisible by 23 ?

   a. 2
   b. 3
   c. 18
   d. 21
   e. None of these

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5. 1397 x 1397 = ?

   a. 1951609
   b. 1981709
   c. 18362619
   d. 2031719
   e. None of these

6. How many of the following numbers are divisible by 132 ?
   264, 396, 462, 792, 968, 2178, 5184, 6336

   a. 4
   b. 5
   c. 6
   d. 7

---

7. (935421 x 625) = ?

   a. 575648125
   b. 584638125
   c. 584649125
   d. 585628125

---

8. The largest 4 digit number exactly divisible by 88 is:

   a. 9944
   b. 9768
   c. 9988
   d. 8888
   e. None of these

---

9. Which of the following is a prime number ?

   a. 33

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b. 81
c. 93
d. 97

---

10. What is the unit digit in $\{(6374)^{1793} \times (625)^{317} \times (341^{491})\}$?

a. 0
b. 2
c. 3
d. 5

11. 5358 x 51 = ?

a. 273258
b. 273268
c. 273348
d. 273358

---

12. The sum of first five prime numbers is:

a. 11
b. 18
c. 26
d. 28

---

13. The difference of two numbers is 1365. On dividing the larger number by the smaller, we get 6 as quotient and the 15 as remainder. What is the smaller number ?

a. 240
b. 270
c. 295
d. 360

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14. $(12)^3 \times 6^4 \div 432 = ?$

a. 5184
b. 5060
c. 5148
d. 5084
e. None of these

---

15. 72519 x 9999 = ?

a. 725117481
b. 674217481
c. 685126481
d. 696217481
e. None of these

16. If the number 517*324 is completely divisible by 3, then the smallest whole number in the place of * will be:

a. 0
b. 1
c. 2
d. None of these

---

17. The smallest 3 digit prime number is:

a. 101
b. 103
c. 109
d. 113

---

18. Which one of the following numbers is exactly divisible by 11?

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a. 235641
b. 245642
c. 315624
d. 415624

---

19. (?) - 19657 - 33994 = 9999

a. 63650
b. 53760
c. 59640
d. 61560
e. None of these

---

20. The sum of first 45 natural numbers is:

a. 1035
b. 1280
c. 2070
d. 2140

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21. Which of the following number is divisible by 24 ?

a. 35718
b. 63810
c. 537804
d. 3125736

---

22. (?) + 3699 + 1985 - 2047 = 31111

a. 34748
b. 27474
c. 30154
d. 27574
e. None of these

---

23. If the number 481 * 673 is completely divisible by 9, then the smallest whole number in place of * will be:

a. 2
b. 5
c. 6
d. 7
e. None of these

---

24. The difference between the local value and the face value of 7 in the numeral 32675149 is

a. 75142
b. 64851
c. 5149
d. 69993
e. None of these

---

25. The difference between a positive proper fraction and its reciprocal is 9/20. The fraction is:

a. 3/5
b. 3/10
c. 4/5
d. 4/3

---

26. On dividing a number by 56, we get 29 as remainder. On dividing the same number by 8, what will be the remainder ?

a. 4

b. 5
c. 6
d. 7

---

27. If n is a natural number, then $(6n^2 + 6n)$ is always divisible by:

a. 6 only
b. 6 and 12 both
c. 12 only
d. by 18 only

---

28. 107 x 107 + 93 x 93 = ?

a. 19578
b. 19418
c. 20098
d. 21908
e. None of these

---

29. What will be remainder when $(67^{67} + 67)$ is divided by 68 ?

a. 1
b. 63
c. 66
d. 67

---

30. On dividing a number by 5, we get 3 as remainder. What will the remainder when the square of the this number is divided by 5 ?

a. 0
b. 1
c. 2
d. 4

DAY_04

DATE : 26-09-2026

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TIME : 6:30 P.M. TO 8 : 30 P.M.

TOPIC : SIMPLE INTEREST

1. A sum of money at simple interest amounts to Rs. 815 in 3 years and to Rs. 854 in 4 years. The sum is:

   a. Rs. 650
   b. Rs. 690
   c. Rs. 698
   d. Rs. 700

---

2. Mr. Thomas invested an amount of Rs. 13,900 divided in two different schemes A and B at the simple interest rate of 14% p.a. and 11% p.a. respectively. If the total amount of simple interest earned in 2 years be Rs. 3508, what was the amount invested in Scheme B?

   a. Rs. 6400
   b. Rs. 6500
   c. Rs. 7200
   d. Rs. 7500
   e. None of these

---

3. A sum fetched a total simple interest of Rs. 4016.25 at the rate of 9 p.c.p.a. in 5 years. What is the sum?

   a. Rs. 4462.50
   b. Rs. 8032.50
   c. Rs. 8900
   d. Rs. 8925
   e. None of these

---

4. How much time will it take for an amount of Rs. 450 to yield Rs. 81 as interest at 4.5% per annum of simple interest?

   a. 3.5 years

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b. 4 years
c. 4.5 years
d. 5 years

---

5. Reena took a loan of Rs. 1200 with simple interest for as many years as the rate of interest. If she paid Rs. 432 as interest at the end of the loan period, what was the rate of interest?

   a. 3.6
   b. 6
   c. 18
   d. Cannot be determined
   e. None of these

---

6. A sum of Rs. 12,500 amounts to Rs. 15,500 in 4 years at the rate of simple interest. What is the rate of interest?

   a. 3%
   b. 4%
   c. 5%
   d. 6%
   e. None of these

---

7. An automobile financier claims to be lending money at simple interest, but he includes the interest every six months for calculating the principal. If he is charging an interest of 10%, the effective rate of interest becomes:
   a. 10%
   b. 10.25%
   c. 10.5%
   d. None of these

---

8. A lent Rs. 5000 to B for 2 years and Rs. 3000 to C for 4 years on simple interest at the same rate of interest and received Rs. 2200 in all from both of them as interest. The rate of interest per annum is:

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a. 5%
b. 7%
c. 57/8
d. 10%

---

9. A sum of Rs. 725 is lent in the beginning of a year at a certain rate of interest. After 8 months, a sum of Rs. 362.50 more is lent but at the rate twice the former. At the end of the year, Rs. 33.50 is earned as interest from both the loans. What was the original rate of interest?

   a. 3.6%
   b. 4.5%
   c. 5%
   d. 6%
   e. None of these

---

10. A man took loan from a bank at the rate of 12% p.a. simple interest. After 3 years he had to pay Rs. 5400 interest only for the period. The principal amount borrowed by him was:

    a. Rs. 2000
    b. Rs. 10,000
    c. Rs. 15,000
    d. Rs. 20,000

---

11. A sum of money amounts to Rs. 9800 after 5 years and Rs. 12005 after 8 years at the same rate of simple interest. The rate of interest per annum is:

    a. 5%
    b. 8%
    c. 12%
    d. 15%

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12. What will be the ratio of simple interest earned by certain amount at the same rate of interest for 6 years and that for 9 years?

    a. 1 : 3
    b. 1 : 4
    c. 2 : 3
    d. Data inadequate
    e. None of these

---

13. A certain amount earns simple interest of Rs. 1750 after 7 years. Had the interest been 2% more, how much more interest would it have earned?

    a. Rs. 35
    b. Rs. 245
    c. Rs. 350
    d. Cannot be determined
    e. None of these

---

14. A person borrows Rs. 5000 for 2 years at 4% p.a. simple interest. He immediately lends it to another person at $6\frac{1}{4}$% p.a for 2 years. Find his gain in the transaction per year.

    a. Rs. 112.50
    b. Rs. 125
    c. Rs. 225
    d. Rs. 167.50

TOPIC : COMPOUND INTEREST

1. A bank offers 5% compound interest calculated on half-yearly basis. A customer deposits Rs. 1600 each on 1st January and 1st July of a year. At the end of the year, the amount he would have gained by way of interest is:

    a. Rs. 120
    b. Rs. 121

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c. Rs. 122
d. Rs. 123

---

2. The difference between simple and compound interests compounded annually on a certain sum of money for 2 years at 4% per annum is Re. 1. The sum (in Rs.) is:

a. 625
b. 630
c. 640
d. 650

---

3. There is 60% increase in an amount in 6 years at simple interest. What will be the compound interest of Rs. 12,000 after 3 years at the same rate?

a. Rs. 2160
b. Rs. 3120
c. Rs. 3972
d. Rs. 6240
e. None of these

---

4. What is the difference between the compound interests on Rs. 5000 for $1\frac{1}{2}$ years at 4% per annum compounded yearly and half-yearly?

a. Rs. 2.04
b. Rs. 3.06
c. Rs. 4.80
d. Rs. 8.30

---

5. The compound interest on Rs. 30,000 at 7% per annum is Rs. 4347. The period (in years) is:

a. 2
b. 5/2

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c. 3
d. 4

---

6. What will be the compound interest on a sum of Rs. 25,000 after 3 years at the rate of 12 p.c.p.a.?

a. Rs. 9000.30
b. Rs. 9720
c. Rs. 10123.20
d. Rs. 10483.20
e. None of these

---

7. At what rate of compound interest per annum will a sum of Rs. 1200 become Rs. 1348.32 in 2 years?

a. 6%
b. 6.5%
c. 7%
d. 7.5%

---

8. The least number of complete years in which a sum of money put out at 20% compound interest will be more than doubled is:

a. 3
b. 4
c. 5
d. 6

---

9. Albert invested an amount of Rs. 8000 in a fixed deposit scheme for 2 years at compound interest rate 5 p.c.p.a. How much amount will Albert get on maturity of the fixed deposit?

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a. Rs. 8600
b. Rs. 8620
c. Rs. 8820
d. None of these

---

10. The effective annual rate of interest corresponding to a nominal rate of 6% per annum payable half-yearly is:

a. 6.06%
b. 6.07%
c. 6.08%
d. 6.09%

---

11. Simple interest on a certain sum of money for 3 years at 8% per annum is half the compound interest on Rs. 4000 for 2 years at 10% per annum. The sum placed on simple interest is:

a. Rs. 1550
b. Rs. 1650
c. Rs. 1750
d. Rs. 2000

---

12. If the simple interest on a sum of money for 2 years at 5% per annum is Rs. 50, what is the compound interest on the same at the same rate and for the same time?

a. Rs. 51.25
b. Rs. 52
c. Rs. 54.25
d. Rs. 60

IET, PUNE

13. The difference between simple interest and compound on Rs. 1200 for one year at 10% per annum reckoned half-yearly is:

    a. Rs. 2.50
    b. Rs. 3
    c. Rs. 3.75
    d. Rs. 4
    e. None of these

---

14. The difference between compound interest and simple interest on an amount of Rs. 15,000 for 2 years is Rs. 96. What is the rate of interest per annum?

    a. 8
    b. 10
    c. 12
    d. Cannot be determined
    e. None of these

---

15. The compound interest on a certain sum for 2 years at 10% per annum is Rs. 525. The simple interest on the same sum for double the time at half the rate percent per annum is:

    a. Rs. 400
    b. Rs. 500
    c. Rs. 600
    d. Rs. 800

DAY_05

DATE : 28-09-2026

TIME : 6:30 P.M. TO 8 : 30 P.M.

IET PUNE

TOPIC : TIME , SPEED AND DISTANCE

1. A person crosses a 600 m long street in 5 minutes. What is his speed in km per hour?

   a. 3.6
   b. 7.2
   c. 8.4
   d. 10

---

2. An aeroplane covers a certain distance at a speed of 240 kmph in 5 hours. To cover the same distance in $1\frac{2}{3}$ hours, it must travel at a speed of:

   a. 300 kmph
   b. 360 kmph
   c. 600 kmph
   d. 720 kmph

---

3. If a person walks at 14 km/hr instead of 10 km/hr, he would have walked 20 km more. The actual distance travelled by him is:

   a. 50 km
   b. 56 km
   c. 70 km
   d. 80 km

---

4. A train can travel 50% faster than a car. Both start from point A at the same time and reach point B 75 kms away from A at the same time. On the way, however, the train lost about 12.5 minutes while stopping at the stations. The speed of the car is:

   a. 100 kmph
   b. 110 kmph
   c. 120 kmph
   d. 130 kmph

---

5. Excluding stoppages, the speed of a bus is 54 kmph and including stoppages, it is 45 kmph. For how many minutes does the bus stop per hour?

   a. 9
   b. 10
   c. 12
   d. 20

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---

6. In a flight of 600 km, an aircraft was slowed down due to bad weather. Its average speed for the trip was reduced by 200 km/hr and the time of flight increased by 30 minutes. The duration of the flight is:

   a. 1 hour
   b. 2 hours
   c. 3 hours
   d. 4 hours

---

7. A man complete a journey in 10 hours. He travels first half of the journey at the rate of 21 km/hr and second half at the rate of 24 km/hr. Find the total journey in km.

   a. 220 km
   b. 224 km
   c. 230 km
   d. 234 km

---

8. The ratio between the speeds of two trains is 7 : 8. If the second train runs 400 km in 4 hours, then the speed of the first train is:

   a. 70 km/hr
   b. 75 km/hr
   c. 84 km/hr
   d. 87.5 km/hr

---

9. A man on tour travels first 160 km at 64 km/hr and the next 160 km at 80 km/hr. The average speed for the first 320 km of the tour is:

   a. 35.55 km/hr
   b. 36 km/hr
   c. 71.11 km/hr
   d. 71 km/hr

---

10. A car travelling with $\frac{5}{7}$ of its actual speed covers 42 km in 1 hr 40 min 48 sec. Find the actual speed of the car.

    a. 125
    b. 25 km/hr
    c. 30 km/hr
    d. 35 km/hr

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11. In covering a distance of 30 km, Abhay takes 2 hours more than Sameer. If Abhay doubles his speed, then he would take 1 hour less than Sameer. Abhay's speed is:

    a. 5 kmph
    b. 6 kmph
    c. 6.25 kmph
    d. 7.5 kmph

---

12. Robert is travelling on his cycle and has calculated to reach point A at 2 P.M. if he travels at 10 kmph, he will reach there at 12 noon if he travels at 15 kmph. At what speed must he travel to reach A at 1 P.M.?

    a. 8 kmph
    b. 11 kmph
    c. 12 kmph
    d. 14 kmph

---

13. It takes eight hours for a 600 km journey, if 120 km is done by train and the rest by car. It takes 20 minutes more, if 200 km is done by train and the rest by car. The ratio of the speed of the train to that of the cars is:

    a. 2 : 3
    b. 3 : 2
    e. 3 : 4
    f. 4 : 3

---

14. A farmer travelled a distance of 61 km in 9 hours. He travelled partly on foot @ 4 km/hr and partly on bicycle @ 9 km/hr. The distance travelled on foot is:

    a. 14 km
    b. 15 km
    c. 16 km
    d. 17 km

---

15. A man covered a certain distance at some speed. Had he moved 3 kmph faster, he would have taken 40 minutes less. If he had moved 2 kmph slower, he would have taken 40 minutes more. The distance (in km) is:

    a. 35
    b. 110
    c. 75/2
    d. 40

IET, PUNE

TOPIC : TIME , WORK AND WAGES

1. A can do a work in 15 days and B in 20 days. If they work on it together for 4 days, then the fraction of the work that is left is :

   a. 1/4
   b. 1/10
   c. 7/15
   d. 8/15

---

2. A can lay railway track between two given stations in 16 days and B can do the same job in 12 days. With help of C, they did the job in 4 days only. Then, C alone can do the job in:

a. 46/5
b. 47/5
c. 48/5
d. 10

---

3. A, B and C can do a piece of work in 20, 30 and 60 days respectively. In how many days can A do the work if he is assisted by B and C on every third day?

   a. 12 days
   b. 15 days
   c. 16 days
   d. 18 days

---

4. A is thrice as good as workman as B and therefore is able to finish a job in 60 days less than B. Working together, they can do it in:

   a. 20 days
   b. 45/2
   c. 25 days
   d. 30 days

---

5. A alone can do a piece of work in 6 days and B alone in 8 days. A and B undertook to do it for Rs. 3200. With the help of C, they completed the work in 3 days. How much is to be paid to C?

IET, PUNE

a. Rs. 375
b. Rs. 400
c. Rs. 600
d. Rs. 800

---

6. If 6 men and 8 boys can do a piece of work in 10 days while 26 men and 48 boys can do the same in 2 days, the time taken by 15 men and 20 boys in doing the same type of work will be:

   a. 4 days
   b. 5 days
   c. 6 days
   d. 7 days

---

7. A can do a piece of work in 4 hours; B and C together can do it in 3 hours, while A and C together can do it in 2 hours. How long will B alone take to do it?

   a. 8 hours
   b. 10 hours
   c. 12 hours
   d. 24 hours

---

8. A can do a certain work in the same time in which B and C together can do it. If A and B together could do it in 10 days and C alone in 50 days, then B alone could do it in:

   a. 15 days
   b. 20 days
   c. 25 days
   d. 30 days

---

9. A does 80% of a work in 20 days. He then calls in B and they together finish the remaining work in 3 days. How long B alone would take to do the whole work?

   a. 23 days
   b. 37 days
   c. $37\frac{1}{2}$
   d. 40 days

---

10. A machine P can print one lakh books in 8 hours, machine Q can print the same number of books in 10 hours while machine R can print them in 12 hours. All the machines are started at 9 A.M. while machine P is closed at 11 A.M. and the

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remaining two machines complete work. Approximately at what time will the work (to print one lakh books) be finished ?

A. 11:30 A.M.
B. 12 noon
C. 12:30 P.M.
D. 1:00 P.M.

---

11. A can finish a work in 18 days and B can do the same work in 15 days. B worked for 10 days and left the job. In how many days, A alone can finish the remaining work

    a. 5
    b. 11/2
    c. 6
    d. 8

---

12. 4 men and 6 women can complete a work in 8 days, while 3 men and 7 women can complete it in 10 days. In how many days will 10 women complete it?

    a. 35
    b. 40
    c. 45
    d. 50

---

13. A and B can together finish a work 30 days. They worked together for 20 days and then B left. After another 20 days, A finished the remaining work. In how many days A alone can finish the work?

    a. 40
    b. 50
    c. 54
    d. 60

---

14. P can complete a work in 12 days working 8 hours a day. Q can complete the same work in 8 days working 10 hours a day. If both P and Q work together, working 8 hours a day, in how many days can they complete the work?

    a. 60 / 11
    b. 61 / 11
    c. 71 / 11
    d. 72 / 11

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15. 10 women can complete a work in 7 days and 10 children take 14 days to complete the work. How many days will 5 women and 10 children take to complete the work?

    a. 3
    b. 5
    c. 7
    d. Cannot be determined
    e. None of these

==========

DAY_06

DATE : 29-09-2026

TIME : 6:30 P.M. TO 8 : 30 P.M.

TOPIC : SIMPLIFICATION

1. A man has Rs.480 in the denominations of one-rupee notes, five-rupee notes and ten-rupee notes. The number of notes of each denomination is equal. What is the total number of notes that he has ?

    a. 45
    b. 60
    c. 75
    d. 90

---

2. There are two examinations rooms A and B. If 10 students are sent from A to B, then the number of students in each room is the same. If 20 candidates are sent from B to A, then the number of students in A is double the number of students in B. The number of students in room A is:

    a. 20
    b. 80
    c. 100
    d. 200

---

3. The price of 10 chairs is equal to that of 4 tables. The price of 15 chairs and 2 tables together is Rs. 4000. The total price of 12 chairs and 3 tables is:

    a. Rs. 3500
    b. Rs. 3750
    c. Rs. 3840
    d. Rs. 3900

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4. If $a - b = 3$ and $a^2 + b^2 = 29$, find the value of ab.

   a. 10
   b. 12
   c. 15
   d. 18

---

5. The price of 2 sarees and 4 shirts is Rs. 1600. With the same money one can buy 1 saree and 6 shirts. If one wants to buy 12 shirts, how much shall he have to pay ?

   a. Rs. 1200
   b. Rs. 2400
   c. Rs. 4800
   d. Cannot be determined
   e. None of these

---

6. A sum of Rs. 1360 has been divided among A, B and C such that A gets $\frac{2}{3}$ of what B gets and B gets $\frac{1}{4}$ of what C gets. B's share is:

   a. Rs. 120
   b. Rs. 160
   c. Rs. 240
   d. Rs. 300

---

7. One-third of Rahul's savings in National Savings Certificate is equal to one-half of his savings in Public Provident Fund. If he has Rs. 1,50,000 as total savings, how much has he saved in Public Provident Fund ?

   a. Rs. 30,000
   b. Rs. 50,000
   c. Rs. 60,000
   d. Rs. 90,000

---

8. A fires 5 shots to B's 3 but A kills only once in 3 shots while B kills once in 2 shots. When B has missed 27 times, A has killed:

   a. 30 birds
   b. 60 birds
   c. 72 birds
   d. 90 birds

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9. Eight people are planning to share equally the cost of a rental car. If one person withdraws from the arrangement and the others share equally the entire cost of the car, then the share of each of the remaining persons increased by:

    a. 1/7
    b. 1/8
    c. 1/9
    d. 7/8

---

10. To fill a tank, 25 buckets of water is required. How many buckets of water will be required to fill the same tank if the capacity of the bucket is reduced to two-fifth of its present ?

    a. 10
    b. 35
    c. 62.5
    d. Cannot be determined
    e. None of these

==========

11. In a regular week, there are 5 working days and for each day, the working hours are 8. A man gets Rs. 2.40 per hour for regular work and Rs. 3.20 per hours for overtime. If he earns Rs. 432 in 4 weeks, then how many hours does he work for ?

    a. 160
    b. 175
    c. 180
    d. 195

---

12. Free notebooks were distributed equally among children of a class. The number of notebooks each child got was one-eighth of the number of children. Had the number of children been half, each child would have got 16 notebooks. Total how many notebooks were distributed ?

    a. 256
    b. 432
    c. 512
    d. 640
    e. None of these

---

13. A man has some hens and cows. If the number of heads be 48 and the number of feet equals 140, then the number of hens will be:

    a. 22
    b. 23

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c. 24
d. 26

14.

$$\frac{(469 + 174)^2 - (469 - 174)^2}{(469 \quad 174)} = ?$$

a. 2
b. 4
c. 295
d. 643

---

15. David gets on the elevator at the 11th floor of a building and rides up at the rate of 57 floors per minute. At the same time, Albert gets on an elevator at the 51st floor of the same building and rides down at the rate of 63 floors per minute. If they continue travelling at these rates, then at which floor will their paths cross ?

A. 19
B. 28
C. 30
D. 37

DAY_07

DATE : 01-10-2026

TIME : 6:30 P.M. TO 8 : 30 P.M.

TOPIC : NUMBER SYSTEM

IET, PUNE

1. On dividing a number by 5, we get 3 as remainder. What will the remainder when the square of the this number is divided by 5 ?

A. 0
B. 1
C. 2
D. 4

2.

How many 3-digit numbers are completely divisible 6 ?

A. 149
B. 150
C. 151
D. 166

3.

How many natural numbers are there between 23 and 100 which are exactly divisible by 6 ?

A. 8
B. 11
C. 12
D. 13
E. None of these

4.

How many of the following numbers are divisible by 3 but not by 9 ?
2133, 2343, 3474, 4131, 5286, 5340, 6336, 7347, 8115, 9276

A. 5
B. 6
C. 7
D. None of these

5.

$$\frac{(963 + 476)^2 + (963 - 476)^2}{(963 \times 963 + 476 \times 476)} = ?$$

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A. 1449
B. 497
C. 2
D. 4
E. None of these

6.
How many 3 digit numbers are divisible by 6 in all ?

A. 149
B. 150
C. 151
D. 166

7.

A 3-digit number 4*a*3 is added to another 3-digit number 984 to give a 4-digit number 13*b*7, which is divisible by 11. Then, (*a* + *b*) = ?

A. 10
B. 11
C. 12
D. 15

8.

8597 - ? = 7429 - 4358

A. 5426
B. 5706
C. 5526
D. 5476
E. None of these

9.

The smallest prime number is:

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A. 1
B. 2
C. 3
D. 4

10.

(12345679 x 72) = ?

A. 88888888
B. 888888888
C. 898989898
D. 9999999998

---

11. On dividing a number by 357, we get 39 as remainder. On dividing the same number 17, what will be the remainder ?

A. 0
B. 3
C. 5
D. 11

12.

If the product 4864 x 9 P 2 is divisible by 12, then the value of P is:

A. 2
B. 5
C. 6
D. 8
E. None of these

13.

Which one of the following is the common factor of $(47^{43} + 43^{43})$ and $(47^{47} + 43^{47})$?

A. (47 - 43)
B. (47 + 43)
C. $(47^{43} + 43^{43})$
D. None of these

14.

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-84 x 29 + 365 = ?

A. 2436
B. 2801
C. -2801
D. -2071
E. None of these

15.

A number when divided by 296 leaves 75 as remainder. When the same number is divided by 37, the remainder will be:

A. 1
B. 2
C. 8
D. 11

---

16. In dividing a number by 585, a student employed the method of short division. He divided the number successively by 5, 9 and 13 (factors 585) and got the remainders 4, 8, 12 respectively. If he had divided the number by 585, the remainder would have been

    A. 24
    B. 144
    C. 292
    D. 584

17.

In a division sum, the divisor is 10 times the quotient and 5 times the remainder. If the remainder is 46, what is the dividend ?

A. 4236
B. 4306
C. 4336
D. 5336
E. None of these

18.

4500 x ? = 3375

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A. 2/5
B. ¼
C. 3/5
D. None of these

19.

What smallest number should be added to 4456 so that the sum is completely divisible by 6 ?

A. 4
B. 3
C. 2
D. 1
E. None of these

20.

What least number must be subtracted from 13601, so that the remainder is divisible by 87 ?

A. 23
B. 31
C. 29
D. 37
E. 49

---

21. 476 ** 0 is divisible by both 3 and 11. The non-zero digits in the hundred's and ten's places are respectively:

A. 7 and 4
B. 7 and 5
C. 8 and 5
D. None of these

22.

If the number 97215 * 6 is completely divisible by 11, then the smallest whole number in place of * will be:

A. 3
B. 2
C. 1
D. 5
E. None of these

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23.

$(11^2 + 12^2 + 13^2 + ... + 20^2) = ?$

A. 385
B. 2485
C. 2870
D. 3255

24.

If the number 5 * 2 is divisible by 6, then * = ?

A. 2
B. 3
C. 6
D. 7

25.

Which of the following numbers will completely divide $(49^{15} - 1)$?

A. 8
B. 14
C. 46
D. 50

---

26. If the number 653 *xy* is divisible by 90, then (*x* + *y*) = ?
A. 2
B. 3
C. 4
D. 6

27.

3897 x 999 = ?

A. 3883203
B. 3893103
C. 3639403
D. 3791203
E. None of these

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28.

What is the unit digit in 7^{105} ?

A. 1
B. 5
C. 7
D. 9

29.

A. Which of the following numbers will completely divide $(4^{61} + 4^{62} + 4^{63} + 4^{64})$?

A. 3
B. 10
C. 11
D. 13

30.

106 x 106 - 94 x 94 = ?

A. 2400
B. 2000
C. 1904
D. 1906
E. None of these

TOPIC : PROFIT AND LOSS

1. Alfred buys an old scooter for Rs. 4700 and spends Rs. 800 on its repairs. If he sells the scooter for Rs. 5800, his gain percent is:

   a. 37/7 %
   b. 60/11 %
   c. 10%
   d. 12%

---

2. The cost price of 20 articles is the same as the selling price of *x* articles. If the profit is 25%, then the value of *x* is:

   a. 15
   b. 16

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c. 18
d. 25

---

3. If selling price is doubled, the profit triples. Find the profit percent.

a. 200/3
b. 100
c. 316/3
d. 120

---

4. In a certain store, the profit is 320% of the cost. If the cost increases by 25% but the selling price remains constant, approximately what percentage of the selling price is the profit?

a. 30%
b. 70%
c. 100%
d. 250%

---

5. A vendor bought toffees at 6 for a rupee. How many for a rupee must he sell to gain 20%?

a. 3
b. 4
c. 5
d. 6

---

6. The percentage profit earned by selling an article for Rs. 1920 is equal to the percentage loss incurred by selling the same article for Rs. 1280. At what price should the article be sold to make 25% profit?
a. Rs. 2000
b. Rs. 2200
c. Rs. 2400

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d. Data inadequate

---

7. A shopkeeper expects a gain of 22.5% on his cost price. If in a week, his sale was of Rs. 392, what was his profit?

   a. Rs. 18.20
   b. Rs. 70
   c. Rs. 72
   d. Rs. 88.25

---

8. A man buys a cycle for Rs. 1400 and sells it at a loss of 15%. What is the selling price of the cycle?

   a. Rs. 1090
   b. Rs. 1160
   c. Rs. 1190
   d. Rs. 1202

---

9. Sam purchased 20 dozens of toys at the rate of Rs. 375 per dozen. He sold each one of them at the rate of Rs. 33. What was his percentage profit?

   a. 3.5
   b. 4.5
   c. 5.6
   d. 6.5

---

10. Some articles were bought at 6 articles for Rs. 5 and sold at 5 articles for Rs. 6. Gain percent is:

    a. 30%
    b. 100/3 %
    c. 35%
    d. 44%

---

11. On selling 17 balls at Rs. 720, there is a loss equal to the cost price of 5 balls. The cost price of a ball is:

    a. Rs. 45
    b. Rs. 50

c. Rs. 55
d. Rs. 60

---

12. When a plot is sold for Rs. 18,700, the owner loses 15%. At what price must that plot be sold in order to gain 15%?

a. Rs. 21,000
b. Rs. 22,500
c. Rs. 25,300
d. Rs. 25,800

---

13. 100 oranges are bought at the rate of Rs. 350 and sold at the rate of Rs. 48 per dozen. The percentage of profit or loss is:

a. 100/7 gain
b. 15% gain
c. 100/7 loss
d. 15 % loss

---

14. A shopkeeper sells one transistor for Rs. 840 at a gain of 20% and another for Rs. 960 at a loss of 4%. His total gain or loss percent is:

a. 110/17 loss
b. 110/17 gain
c. 20/3 gain
d. None of these

---

15. A trader mixes 26 kg of rice at Rs. 20 per kg with 30 kg of rice of other variety at Rs. 36 per kg and sells the mixture at Rs. 30 per kg. His profit percent is:

a. No profit, no loss
b. 5%
c. 8%

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d. 10%
e. None of these

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